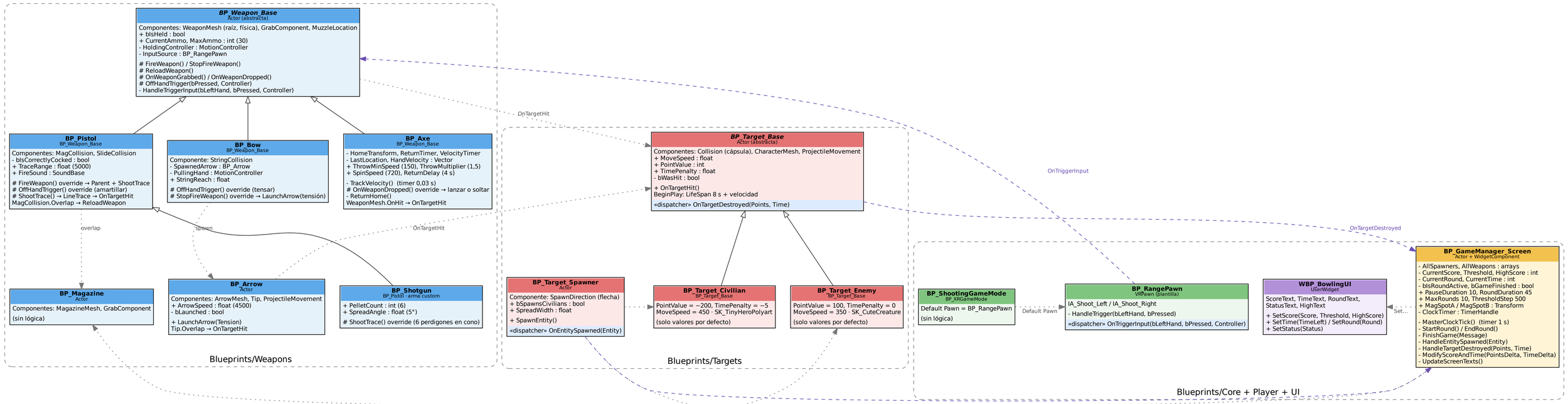
Simulador de Tiro VR (UE 5.7) - Diagrama UML de clases
Sólida + triángulo hueco = herencia · Discontinua morada = Event Dispatcher (emisor → oyente) · Punteada gris = referencia / spawn / llamada
BP_Weapon_Base
Actor (abstracta)
Componentes: WeaponMesh (raíz, física), GrabComponent, MuzzleLocation
+ bIsHeld : bool
+ CurrentAmmo, MaxAmmo : int (30)
- HoldingController : MotionController
- InputSource : BP_RangePawn
FireWeapon() / StopFireWeapon()
ReloadWeapon()
OnWeaponGrabbed() / OnWeaponDropped()
OffHandTrigger(bPressed, Controller)
- HandleTriggerInput(bLeftHand, bPressed, Controller)
BP_Pistol
BP_Weapon_Base
Componentes: MagCollision, SlideCollision
- bIsCorrectlyCocked : bool
+ TraceRange : float (5000)
+ FireSound : SoundBase
# FireWeapon() override → Parent + ShootTrace
# OffHandTrigger() override (amartillar)
# ShootTrace() → LineTrace → OnTargetHit
MagCollision.Overlap → ReloadWeapon
BP_Bow
BP_Weapon_Base
Componente: StringCollision
- SpawnedArrow : BP_Arrow
- PullingHand : MotionController
+ StringReach : float
# OffHandTrigger() override (tensar)
# StopFireWeapon() override → LaunchArrow(tensión)
BP_Axe
BP_Weapon_Base
- HomeTransform, ReturnTimer, VelocityTimer
- LastLocation, HandVelocity : Vector
+ ThrowMinSpeed (150), ThrowMultiplier (1,5)
+ SpinSpeed (720), ReturnDelay (4 s)
- TrackVelocity() (timer 0,03 s)
OnWeaponDropped() override → lanzar o soltar
- ReturnHome()
WeaponMesh.OnHit → OnTargetHit
BP_Magazine
Actor
Componentes: MagazineMesh, GrabComponent
(sin lógica)
BP_Arrow
Actor
Componentes: ArrowMesh, Tip, ProjectileMovement
+ ArrowSpeed : float (4500)
- bLaunched : bool
+ LaunchArrow(Tension)
Tip.Overlap → OnTargetHit
BP_Shotgun
BP_Pistol · arma custom
+ PelletCount : int (6)
+ SpreadAngle : float (5°)
# ShootTrace() override (6 perdigones en cono)
Blueprints/Weapons
BP_Target_Base
Actor (abstracta)
Componentes: Collision (cápsula), CharacterMesh, ProjectileMovement
+ MoveSpeed : float
+ PointValue : int
+ TimePenalty : float
- bWasHit : bool
+ OnTargetHit()
BeginPlay: LifeSpan 8 s + velocidad
«dispatcher» OnTargetDestroyed(Points, Time)
BP_Target_Spawner
Actor
Componente: SpawnDirection (flecha)
+ bSpawnsCivilians : bool
+ SpreadWidth : float
+ SpawnEntity()
«dispatcher» OnEntitySpawned(Entity)
BP_Target_Civilian
BP_Target_Base
PointValue = −200, TimePenalty = −5
MoveSpeed = 450 · SK_TinyHeroPolyart
(solo valores por defecto)
BP_Target_Enemy
BP_Target_Base
PointValue = 100, TimePenalty = 0
MoveSpeed = 350 · SK_CuteCreature
(solo valores por defecto)
Blueprints/Targets
BP_ShootingGameMode
BP_XRGameMode
Default Pawn = BP_RangePawn
(sin lógica)
BP_RangePawn
vrPawn (plantilla)
IA_Shoot_Left / IA_Shoot_Right
- HandleTrigger(bLeftHand, bPressed)
«dispatcher» OnTriggerInput(bLeftHand, bPressed, Controller)
WBP_BowlingUI
UserWidget
ScoreText, TimeText, RoundText,
StatusText, HighText
+ SetScore(Score, Threshold, HighScore)
+ SetTime(TimeLeft) / SetRound(Round)
+ SetStatus(Status)
BP_GameManager_Screen
Actor + WidgetComponent
- AllSpawners, AllWeapons : arrays
- CurrentScore, Threshold, HighScore : int
- CurrentRound, CurrentTime : int
- bIsRoundActive, bGameFinished : bool
+ PauseDuration 10, RoundDuration 45
+ MaxRounds 10, ThresholdStep 500
+ MagSpotA / MagSpotB : Transform
- ClockTimer : TimerHandle
- MasterClockTick() (timer 1 s)
- StartRound() / EndRound()
- FinishGame(Message)
- HandleEntitySpawned(Entity)
- HandleTargetDestroyed(Points, Time)
- ModifyScoreAndTime(PointsDelta, TimeDelta)
- UpdateScreenTexts()
Blueprints/Core + Player + UI
OnTargetHit
OnTargetHit
OnTriggerInput
OnTargetDestroyed
overlap
spawn
Default Pawn
Set...
spawn
spawn
OnEntitySpawned